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Comparison of Matrix Acidizing, Acid Fracturing and Proppant Fracturing for Stimulation of a Carbonate Exploration Prospect in Pakistan

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Abstract

A gas bearing Paleocene to Early Eocene carbonate formation is present in a fold and thrust belt in Pakistan and is a current exploration target. This formation overlies a Cretaceous clastic sandstone formation, which is currently being produced from many wells, providing logging data indicating its thickness over a wide area. The carbonate formation consists of both massive and thin/thick bedded limestones and marlstones. Available data indicates that porosity and permeability are extremely low, and although natural fractures are likely present, their properties are unknown. A study was conducted to predict the potential benefits of three stimulation methods—matrix acidizing, acid fracturing, and proppant fracturing—to determine if any of these methods could lead to the creation of an economic well.

Core from a recently drilled well was subjected to a suite of regular and special core analyses to assist in evaluating the formation properties and providing data for stimulation design. A prospective location has been identified in a neighboring field where the carbonate formation is known to be significantly thicker, presenting a more attractive target for an appraisal well. A study was performed using industry-standard simulation tools to determine what kind of stimulation effect (negative skin or effective fracture half-length and conductivity) could be achieved. These results were entered into a single-well, single-layer, single phase reservoir simulation model to obtain production forecasts, which were then compared with a base-case unstimulated vertical well.

The study results indicate that based on the currently available data, the potential for matrix acidizing is not sufficient to make an economic well. Acid fracturing is a potential strategy, although significant uncertainties remain regarding the long-term behavior of an acid fracture in this type of formation. Propped fracturing delivers the largest stimulation benefit with the least uncertainty about the long-term performance but carries significant placement risks based on experience in the underlying sandstone formation.

This work presents an investigation into a new potential gas-producing formation in Pakistan based on log and core data, and it compares the potential benefits of three different stimulation methods. The results of this study are beneficial for other operators who are considering exploration or appraisal wells in this carbonate formation or other similar unconventional targets.

Introduction

Pakistan has a long history of gas production from both carbonate and sandstone formations. Currently producing carbonate formations depend on naturally fractures to enable economic production because of low matrix permeability. The current exploration target has matrix porosity and permeability much lower than any producing fractured carbonate reservoir in Pakistan. The extent, frequency and permeability of natural fractures in this formation are highly uncertain. As part of the exploration well planning process, it was necessary to determine what type of well stimulation was most likely to deliver the best production rate and had the best chance of operational success. The results of this study have been used as part of the concept selection for the exploration well planning process, since the type of stimulation method chosen affects the completion options for the well.

Location and Geological Setting

The target formation is a Paleocene/Eocene carbonate reservoir situated within the southern Pakistan Fold and Thrust Belt. The structural style in this area is comprised primarily of compression along with strike-slip components having maximum stress direction roughly NW - SE. The Paleocene/Eocene carbonate is composed of three distinct lithological units. The upper unit primarily consists of argillaceous or marly limestone, which is occasionally fractured. The middle unit is characterized by layers of argillaceous marlstone, claystone and/or shale, while the lower unit is comprised of fractured limestone. Stratigraphically, the Paleogene carbonate is underlain by Paleocene clastic deposits and overlain by Eocene shales, which is candidate seal rock. The source rocks for the target reservoir are claystones/shales of Cretaceous and Paleocene age within the area of interest. The target formation exhibits significant variability in thickness, ranging from 88 to 241 meters. A cross-section of relevant study wells can be seen in Figure 1.

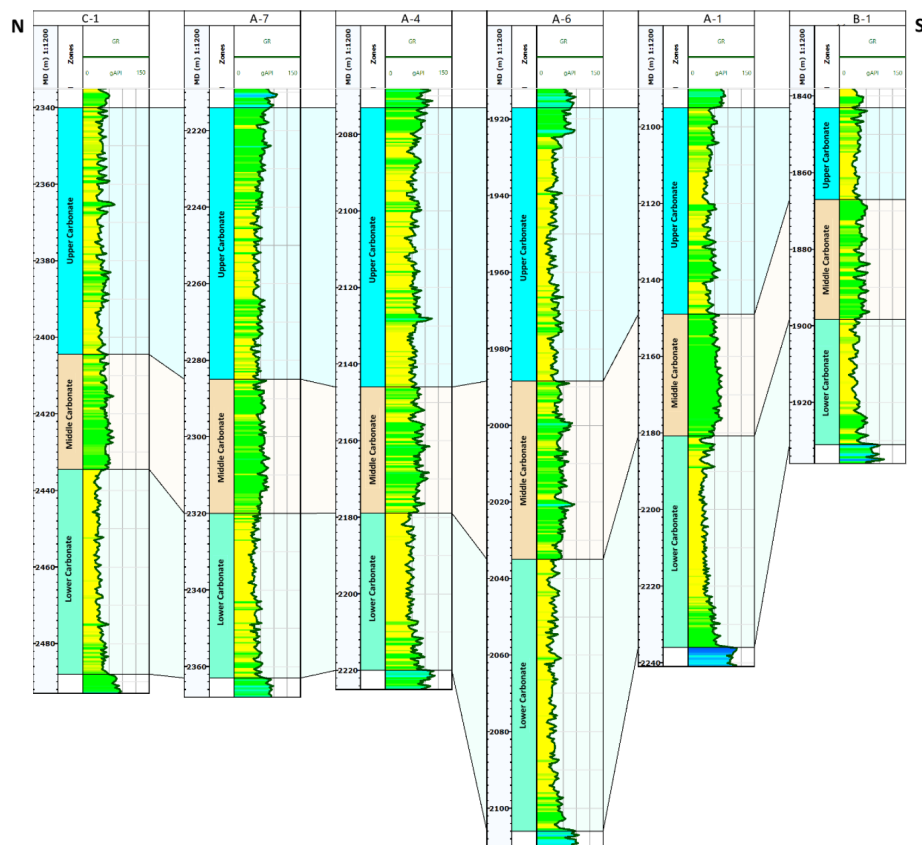


Figure 1—Well correlation across fields A, B and C.

Data from Analog Wells

In this study, data from analogue wells B-1 and A-6 was used that encompassed a comprehensive suite of core and cuttings analyses, including Routine Core Analysis (RCA), petrography, rock mechanics, and formation damage assessments. Additionally, wireline log data, comprising both conventional logs and image logs, was integrated into the evaluation. A potential stimulation candidature and best stimulation type within the targeted carbonate formation was identified in a new well situated in close proximity to well A-6. Petrophysical parameters for this candidate were inferred using data from analogue well A-6, with supplementary support from well B-1. To address subsurface uncertainties and mitigate risk, three reservoir property scenarios—Best, Base, and Worst cases—were established based on variations in porosity and permeability. The best-case scenario was derived from log data and petrophysical analysis, while the base and worst cases were constructed by systematically reducing matrix porosity and estimating corresponding permeability reductions for lower porosity intervals.

Building Simulation Model for Stimulation Designs

Data was used from two existing wells that had penetrated the target (exploration) formation while being drilled to the underlying clastic formation that is currently being produced. One well (well B-1) had an extensive data set, consisting of core & cuttings and various log data (conventional logs, image logs, petrography & rock mechanics). The second well (well A-6) was located close to the planned location for a new dedicated exploration well, but only basic log data was taken in this well. There was no flow testing of the target formation performed in either of these wells. To build the models required for simulating the different types of stimulations, the data from these two wells would have to be combined and extrapolated or interpolated as necessary. This will be explained in more detail in the subsequent paragraphs.

A machine-learning-based approach was used to predict the potential presence and effects of natural fractures by generating estimates of fracture porosity, effective porosity (including fracture porosity) and permeability (Figure 2). The machine-learning approach was trained and calibrated on four different wells with log data in the target formation. The final output from this process was a log including the newly generated properties for the closest offset well to where the new exploration well was planned to be drilled. Based on the large scale structural\geological model, it was estimated that the reservoir depth and thickness would be the same for the new location as for this offset well, so all the models were built using this new log data.

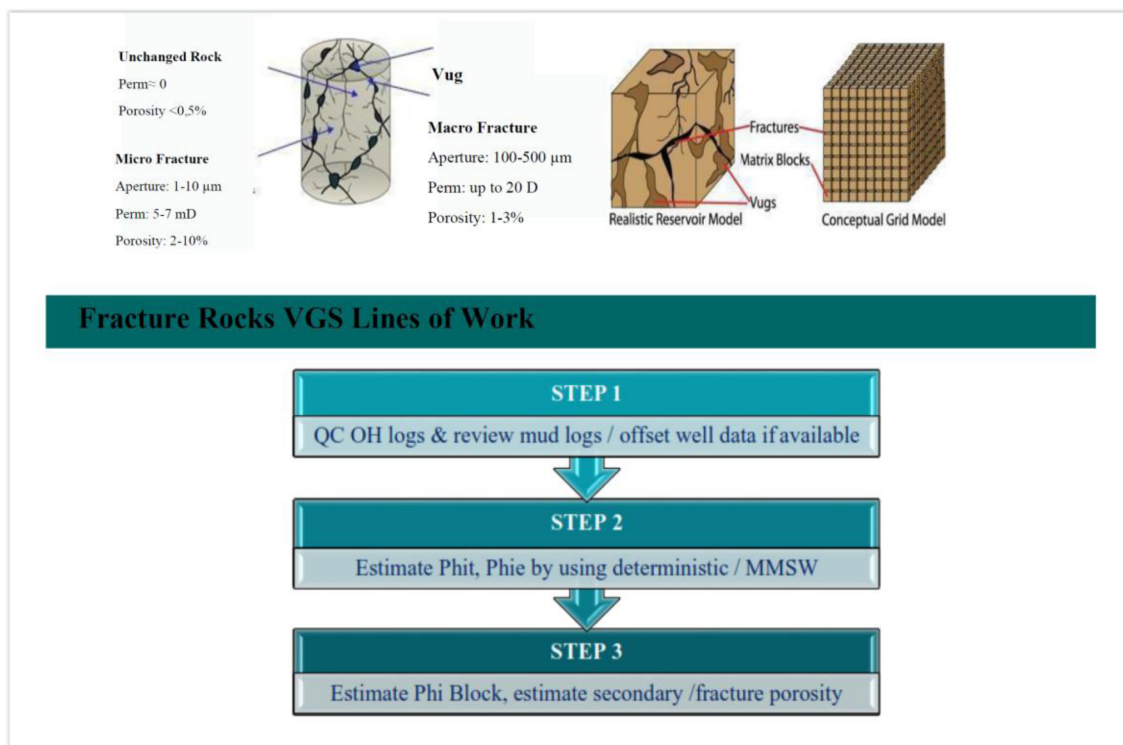


Figure 2—Methodology to estimate porosity and permeability in fractured low permeability reservoir.

Reservoir Property Modeling

Reservoir properties are needed to create a multi-layer model with a fine enough layering to perform simulations of all three stimulation methods: matrix acidizing, acid fracturing, and propped fracturing. A single reservoir model was created that was used for all three types of simulation since a commercial software package was used from a single vendor. The reservoir properties for this model needed to be estimated\generated for the conditions that exist during stimulation injections of a liquid rather than under producing conditions with gas. For the production forecasting, the reservoir properties need to be averaged for a single-layer model and various adjustments made to account for the potential variations in reservoir quality and natural fracture properties. These are discussed later in the section on production forecasting. Real drilling data across the target reservoir gives a bottomhole static temperature (BHST) of 187°F and a pore pressure gradient is 0.58 psi/ft resulting in a reservoir pressure around 3,700 psi at the study reservoir depth.

The target reservoir is split between upper and lower intervals for this study. For effective fracture stimulation of the targeted pay intervals, both layers need to be stimulated in separate treatments for many reasons. For effective zonal coverage, it will take two fracture treatments to cover the targeted pay intervals that are spread across the large gross reservoir thickness of 150 to 200 meters. The ability to cover and connect both the upper and lower intervals is further compromised by the presence of a 30 to 50 meter thick middle section consisting of layers of argillaceous marlstone interbedded with claystone/shale layers. And finally, a limited entry approach to attempt to stimulate both intervals at the same time is considered to have too many uncertainties in an exploration well context.

The layering selection for the reservoir model was driven by the volume of clay log channel, V_{cl} , as the target intervals consist of layers of carbonate-rich rock with varying amounts of clays and silt. Table 1 lists the model rock types and their respective layering criteria. Other log-based petrophysical properties, such as porosity and water saturation, were also considered to finalize the detailed layering. Intervals with open natural fractures have their own lithology designation as this may affect stimulation treatment strategy and

production forecasting. Open natural fractures were only found in the harder, more competent limestone and dirty limestone layers.

Table 1—Rock types and layering criteria.

Increasing Clay Content ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓	Rock Type	Volume of Clay (Vcl)	Natural Fracture Porosity	Increasing Carbonate Content ↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑
	Clean Limestone	< 0.1	-	
	Clean Limestone + Natural Fractures	< 0.1	> 0.001	
	Dirty Limestone	0.1 – 0.2	-	
	Dirty Limestone + Natural Fractures	0.1 – 0.2	> 0.001	
	Argillaceous Limestone	0.2 – 0.3	-	
	Marlstone	> 0.3	-	
	Eocene Shale (Over-burden)	> 0.3	-	
	Paleocene Claystone (Under-burden)	> 0.3	-	

With the reservoir layering in place, key petrophysical properties including porosity and water saturation, were assigned from the Computer Processed Interpretation (CPI) log. The water saturation curve, S_w , represents only the free water in this application, not the bound water. Therefore, effective porosity was used for a more reliable calculation of hydrocarbon filled pore space and because it was available across all data sets, including those generated by machine learning. Pay intervals were then chosen based on criteria of being clean or dirty limestone layers with a water saturation less than 60% and an effective porosity greater than 1.0%.

The reservoir layering for the upper and lower sections of the target formation, based on well A-6, are shown in Figure 3 and Figure 4, respectively. The various lithologies are shown in the column "Rock Type" – the pay intervals in the target formation are labelled as either clean or dirty limestone or fractured limestone to identify the areas that are potentially naturally fractured. The only difference between the two limestone lithologies is that the fractured limestone layers have a higher estimated permeability (a synthetic parameter coming from the log analysis described in the previous section). The non-pay layers are some dirty limestone layers, along with the argillaceous limestone and marlstone layers.

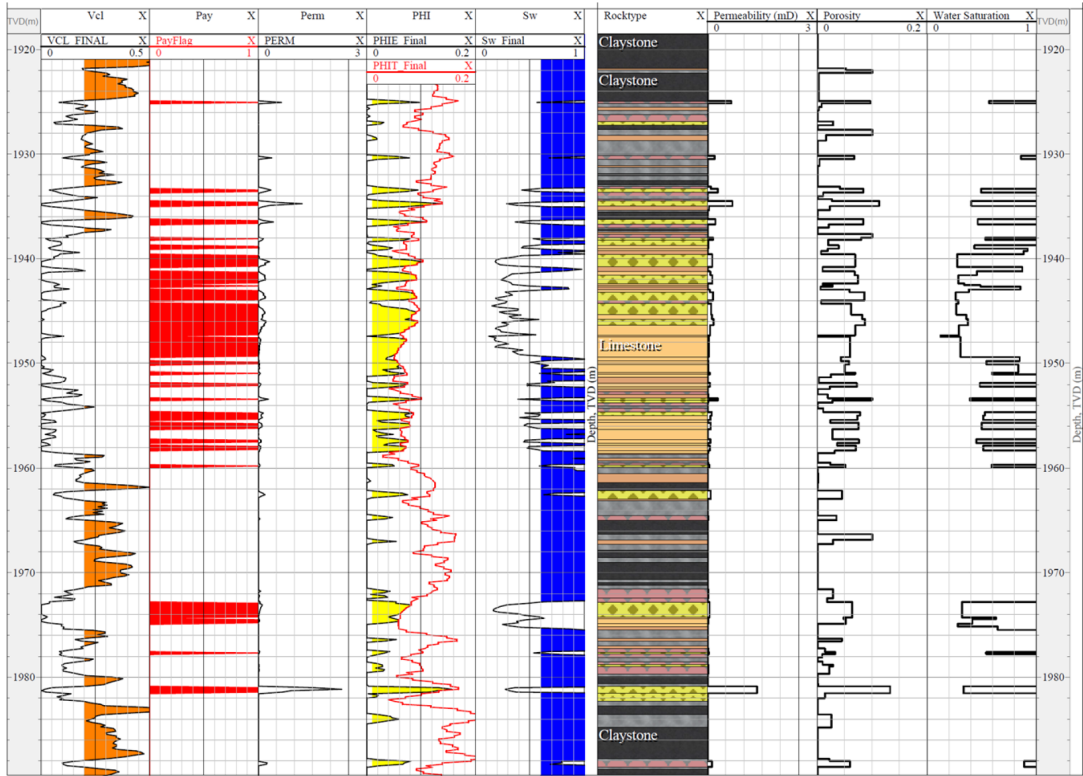


Figure 3—Reservoir model and layering for upper section of target formation.

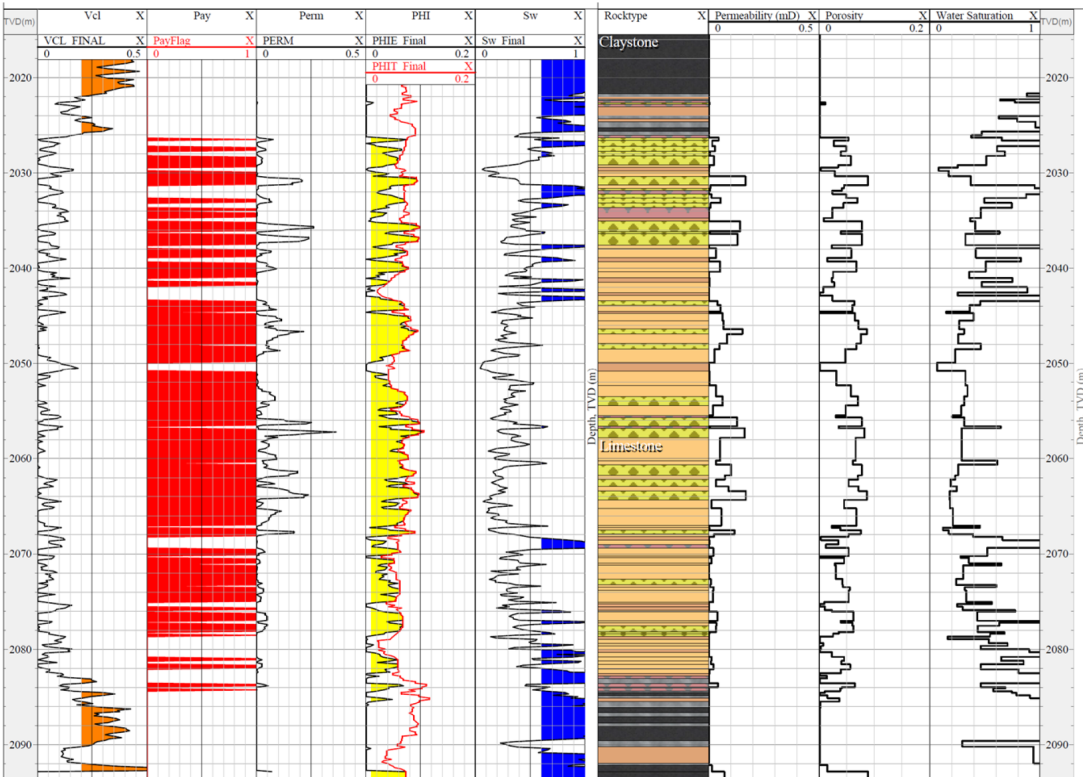


Figure 4—Reservoir model and layering for lower section of target formation.

A summary of the petrophysical analysis based on the synthetic log properties for both the upper (1-3) and lower (4-6) target intervals is shown in Table 2. Between the upper and lower target intervals there

is a claystone\shale layer, which means that for fracture stimulation, they would need to be stimulated in separate treatments. A limited entry approach to try fracture both intervals at the same time is considered to have too many uncertainties in an exploration well context.

Table 2—Petrophysical summary of upper and lower target formations from offset well.

Petrophysical Summary (Sub Zones)												
	Flag Name	Top	Bottom	unit	Gross (m)	Net (m)	Net to Gross	Av_Shale Volume (m3/m3)	Av_Porosity (m3/m3)	Av_Water Saturation (m3/m3)	Av_PERM (mD)	Av_PHIFRAC (m3/m3)
1	ROCK	1917	1995	m	78	41	0.53	0.076	0.041	0.571	0.075	0.012
2	RES	1917	1995	m	78	29	0.37	0.048	0.057	0.565	0.094	0.012
3	PAY	1917	1995	m	78	16	0.21	0.016	0.072	0.363	0.14	0.01
4	ROCK	2035	2106	m	71	64	0.90	0.045	0.046	0.401	0.038	0.006
5	RES	2035	2106	m	71	54	0.77	0.033	0.054	0.4	0.045	0.006
6	PAY	2035	2106	m	71	43	0.61	0.024	0.059	0.326	0.055	0.005

The estimated porosities from the synthetic log for well A-6 are all based on effective porosity (PHIE), so the saturation values are the mobile (ignoring bound water) water saturations. These estimated porosity values for the PAY are around 6%, which is a significantly higher value than what was calculated from the actual logs well B-1, which was in the range of 1-3%. Due to this difference, a range of porosity values have been chosen for the production forecasting to capture this uncertainty. Similarly, the estimated permeability values from the synthetic log result in a much higher kh of 4.9 mD-m (including both upper and lower intervals) which is an order of magnitude larger than the permeability measured in the core testing of well B-1. A range of permeability values to cover this uncertainty is used, with the best case coming from well A-6 synthetic logs and the other cases based on the average porosity value and the existing poro-perm relationship. These average properties are shown later in the paper.

Rock Mechanics and Stress Modeling

Rock mechanical properties and in-situ stress are important parts of any fracture modelling exercise for both propped and acid fracturing. No representative core data was available, so data was taken from available drilling geomechanics data/studies on wells in the field that have penetrated the target formation and calculated from sonic logs. Calibration and checks on data calculated from sonic logs were performed by comparing results to regional core/published data and by comparing calculated results across currently producing intervals to known data points from cores and fracture treatments. While not an ideal calibration, this process did at least provide some assurance that calculated and assumed rock mechanical and stress data are within a reasonable range for initial fracture designs to be representative. Reservoir model rock mechanical property and minimum in-situ stress profiles can be seen in [Figure 5](#).

Poisson's ratio and Young's modulus were calculated from sonic log data using standard elastic relationships, with shear and compressional slowness (DTS and DTC) from the dipole sonic log and bulk density from the density log. Young's modulus is important to modelling fracture dimensions and this exercise resulted in a range from below 3-million psi in the high-clay containing marlstone layers to over 9-million psi in the clean, hard limestone layers of the lower target interval.

The minimum in-situ stress, S_{hmin} , and the associated vertical stress contrast remain significant unknowns. These are important in fracture modelling as they affect many model results, including pressure, hydraulic width and height containment. Calculating stress from sonic logs is an inexact exercise. The approach to calculating/estimating S_{hmin} for this project was used to provide an initial estimate for the design exercise

and to capture a realistic range of outcomes that can be used to understand their effect on fracture modelling outcomes.

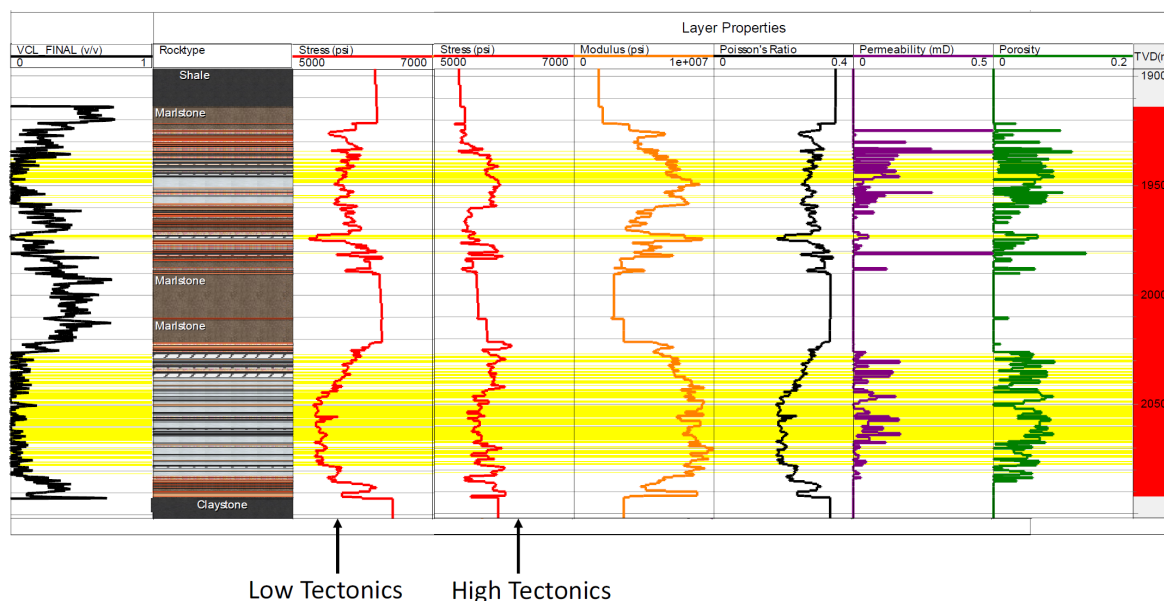


Figure 5—Reservoir model rock mechanical property and minimum in-situ stress profiles.

The average S_{hmin} gradient across the entire target interval has been calculated to be around 0.87 psi/ft, as predicted by drilling geomechanics reports and drilling results, which is higher than most formations at just under 2,000 mTVDss. This is likely due to two unique drivers. The first is the over-pressured pore pressure gradient of 0.58 psi/ft, which will drive up the S_{hmin} across the entire interval and may reduce the stress contrast between shaly layers (traditional barriers when fracturing) and limestone pay layers. The second reason is the strike-slip stress regime observed in the area due to strong tectonics, which is no surprise due to the proximity to significant mountain ranges. This is also expected to increase the closure stress across the entire target interval. However, if it is driving up the stress in the S_{hmin} direction, tectonics may preferentially increase stress in the harder, more consolidated, higher Young's modulus limestone pay layers than softer, lower Young's modulus non-pay layers. If tectonics play a significant role, there may be limited to no stress contrast between limestone pay layers and bounding marlstone/claystone/shale layers.

To capture a range of outcomes for the stress profile, two S_{hmin} stress profile scenarios were considered for fracture modelling. The first, referred to as the "low tectonics" scenario, assumes that a stress contrast does exist between lower stressed limestone pay intervals and higher stressed claystone/marlstone/shale barriers. The second scenario, referred to as the "high tectonics" scenario, assumes there is limited to no stress contrast across the target interval, meaning limestone pay intervals are on average around equal to the claystone/marlstone/shale barriers. This second scenario will result in less fracture height containment which may even affect fracture placement.

The low tectonics scenario calculated a lithological stress component from the Eaton equation and assumed a constant tectonic component of 1,140 psi that was added across the entire interval, which set the average pay stress gradient to the aforementioned 0.87 psi/ft. The high tectonic scenario used the same lithological component from the Eaton equation, but calculated a tectonic component for each layer from tectonic strain using elastic theory. This calculation relies on a relationship between Young's modulus and Poisson's ratio, but is driven by the Young's modulus meaning that the harder, high Young's modulus limestone pay layers will see a higher tectonic stress component. A common tectonic strain value was used

as the calibration variable to adjust the average pay stress gradient to the same 0.87 psi/ft. A comparison of the two S_{hmin} curves can be seen in [Figure 5](#).

An additional height growth confinement mechanism, in addition to stress contrast, included in the fracture modelling is the composite layering effect, or CLE (Warpinski et al., 1998). CLE is used to capture additional resistance to hydraulic fracture height growth across material or layer interfaces as opposed to contrasts in mechanical properties in different layers. For this study, values were based on regional experience and real data points from fracturing underlying reservoirs to the target formation. CLE values range from 1 to 2 in clean/dirty limestone, where little resistance to height growth is expected, up to 5 to 10 in the argillaceous limestone and marlstone where the clay content increases. A regional value for shales of 20 was used for the overburden and under-burden, where moderate to strong resistance to fracture height growth is expected.

Stimulation Design

With the reservoir model built, stimulation designs prepared for all three stimulation methods to be evaluated: matrix acidizing, acid fracturing and proppant fracturing. All modelling was performed in an industry-proven commercial software package. The primary goal of the design exercise was to model practical base case stimulation designs that can be considered realistic and/or conservative to pump and place in this formation. Key challenges, uncertainties and risks were also highlighted for each treatment. The upper and lower intervals are treated separately in all cases. And for all fracturing cases, both propped and acid, both the low and high tectonics cases are modelled. For the matrix acidizing case, the stress profile is not relevant, only the minimum in-situ stress is relevant so the bottomhole injection pressure can remain below the value to avoid breaking down the formation and fracturing.

Proppant Fracturing

Conventional propped fracturing treatments are considered in this exercise. Treatments were designed to target after-closure average proppant concentration in the fracture above 1.5 lb/ft² using a 25-mesh lightweight ceramic proppant (LWP). Ideally, at least 2.0 lb/ft² would be targeted for improved post-frac clean-up of crosslinked gel in this tight gas reservoir, but placement at average concentrations much greater than 1.0 lb/ft² are expected to be challenging in this formation consisting of hard, high Young's modulus limestone in strike-slip stress regime where hydraulic width while pumping may be challenging due to uncertain fracture height containment.

To make designs more conservative and improve proppant placement, larger pad stage volumes were included that were not just based on fluid efficiency alone. Also, most of the proppant was placed before attempting a 6 ppa proppant concentration stage. With initial exploration wells planned to be vertical, the perforation interval (and fracture initiation interval) was chosen as a 3 meter interval towards the middle of the target interval in a low stress area with a high concentration of limestone pay layers. The perforation interval was limited to 3 meters to mitigate development of multiple fractures and fracture complexity. The fracture initiation interval is highlighted by a red bar on the depth axis on all fracture placement result figures.

A low-to-moderate amount of fracture complexity and extra leak-off due to the possible opening of natural fractures while pumping was included in the model to account for potential fracture complexity in this hard, tight rock and additional leak-off. Higher-than-normal proppant drag was considered, assuming hydraulic fracture width will be low. Finally, 500 psi of tortuosity and 500 psi of perforation friction are included based on observations from fracturing other intervals in the area.

A base case treatment schedule representative of all modelling results presented can be seen in [Table 3](#). This design has a clean pad volume around 30%, more than the high fluid efficiency of this tight formation dictates, to aid in generating sufficient hydraulic width for easier proppant admittance and placement. The pad volume has been adjusted across various scenarios to maintain a consistent net pressure build during

placement. The treatment also includes long 4 and 5 ppa slurry stages for improved decision making and real-time adjustments, with most of the proppant, over 80%, placed prior to starting the 6 ppa slurry stage. A small 7 ppa stage is included, if possible, to improve near-wellbore fracture conductivity, which should help with fracture cleanup and mitigating any effects from non-Darcy, multiphase, convergent flow. Diagnostic injections need to be planned in advance on any exploration well and 100-mesh sand available for proppant slugs is recommended.

Table 3—Propped fracturing treatment schedule and material totals.

Stage #	Stage Type	Fluid Type	Clean Volume (gal)	Prop Conc (ppg)	Stage Prop. (klbs)	Slurry Rate (bpm)	Proppant Type
1	Main frac pad	Xlinked Gel 30#	20,000	0.00	0.0	20.00	
2	Main frac slurry	Xlinked Gel 30#	5,000	1.00	5.0	20.00	LWP LD25
3	Main frac slurry	Xlinked Gel 30#	5,000	2.00	10.0	20.00	LWP LD25
4	Main frac slurry	Xlinked Gel 30#	5,000	3.00	15.0	20.00	LWP LD25
5	Main frac slurry	Xlinked Gel 30#	10,000	4.00	40.0	20.00	LWP LD25
6	Main frac slurry	Xlinked Gel 30#	10,000	5.00	50.0	20.00	LWP LD25
7	Main frac slurry	Xlinked Gel 30#	4,000	6.00	24.0	20.00	LWP LD25
8	Main frac slurry	Xlinked Gel 30#	1,000	7.00	7.0	20.00	LWP LD25
9	Main frac flush	Linear Gel 30#	3,800 / 4,400	0.00	0.0	20.00	

Material	Quantity	Units
Xlinked Gel 30#	1,428	bbbls
Linear Gel 30#	105	bbbls
Lightweight Ceramic Proppant LD25	150	klbs

A presentative treating plot for a propped frac in the upper interval (low tectonics case) can be seen in Figure 6. This treatment is pumped at 20 bpm and sees the highest surface treating pressure of all scenarios at 6,150 psi. While the net pressure is relatively high in pad, only a modest net pressure gain of 370 psi is observed while placing proppant. The lower interval treating plot for the high tectonics case seen in Figure 7 shows a slightly lower surface treating pressure and pad net pressure but has a slightly more aggressive net pressure rise while placing the proppant of 580 psi. The treating plot profiles indicate that propped fracturing placement may be possible in this formation.

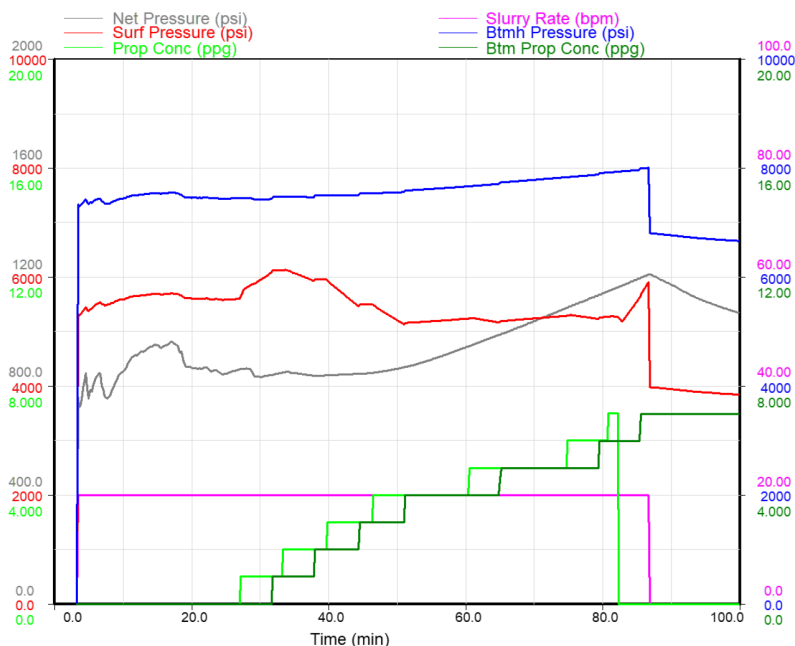


Figure 6—Propped fracture treating plot for upper interval low tectonics case.

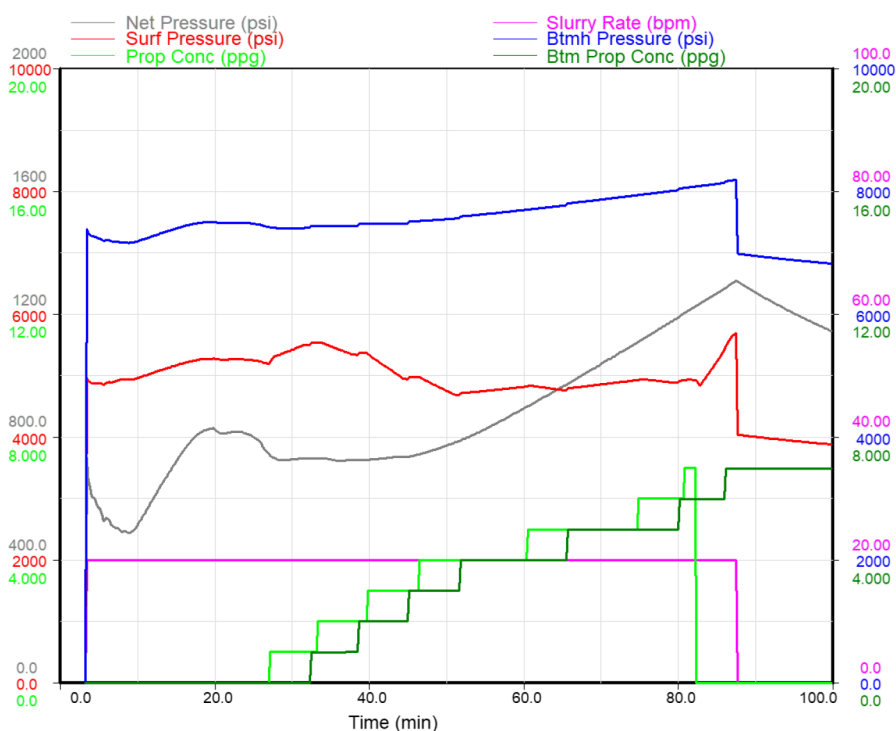


Figure 7—Propped fracture treating plot for lower interval high tectonics case.

Propped fracture placement results for these same treatments can be seen in Figure 8 and Figure 9, respectively. A summary of all placement results can be seen in Table 4. The most important points to highlight are that these more conservative treatments are modelled as staying contained with the target intervals in all cases and that the after-closure proppant concentration target of 1.5 lb/ft² is exceeded in all cases. The treatments placed in high tectonics cases, where there is less stress contrast providing fracture containment, do tend to result in more propped height and less propped length. But no runaway height growth is observed and all cases results in full, or near-full, vertical zonal coverage of the pay intervals. The tips of the fracture where there is lower proppant concentration are unlikely to clean-up and contribute to production. This is addressed in the production forecasting section. An aggressive breaker schedule and gel loading are recommended to promote fracture clean-up of the crosslinked gel.

These results provide tepid optimism that propped fractures can be placed in this formation. The practical reality is that a cautious approach will need to be taken for any propped fracture treatment as there exists a real possibility that proppant placement remains challenging due to insufficient hydraulic width results in a near-wellbore screenout or early proppant bridging of the proppant in the formation. An improved understanding of the stress contrast, and any CLE effect that may also help with fracture containment, will help quantify the risk. While pumping, any fracture growth out of zone, either controlled or uncontrolled, is likely to result in a reduction in fracture width. The other placement risk comes from the potential opening of natural fractures/fissures will pumping. If the increase in leak-off is too high and cannot be controlled, proppant placement will also become more challenging. All of this is the reality of attempting to place proppant in the hard, tight limestone formations in of the Middle East.

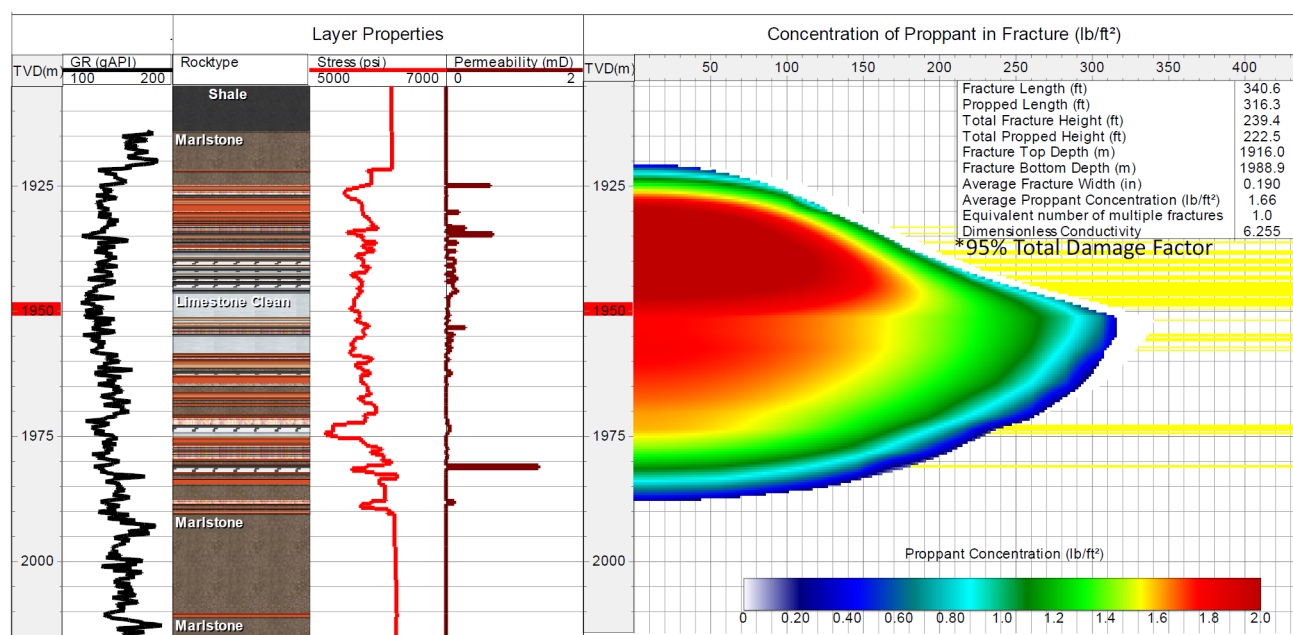


Figure 8—Propped fracture placement results for upper interval low tectonics case.

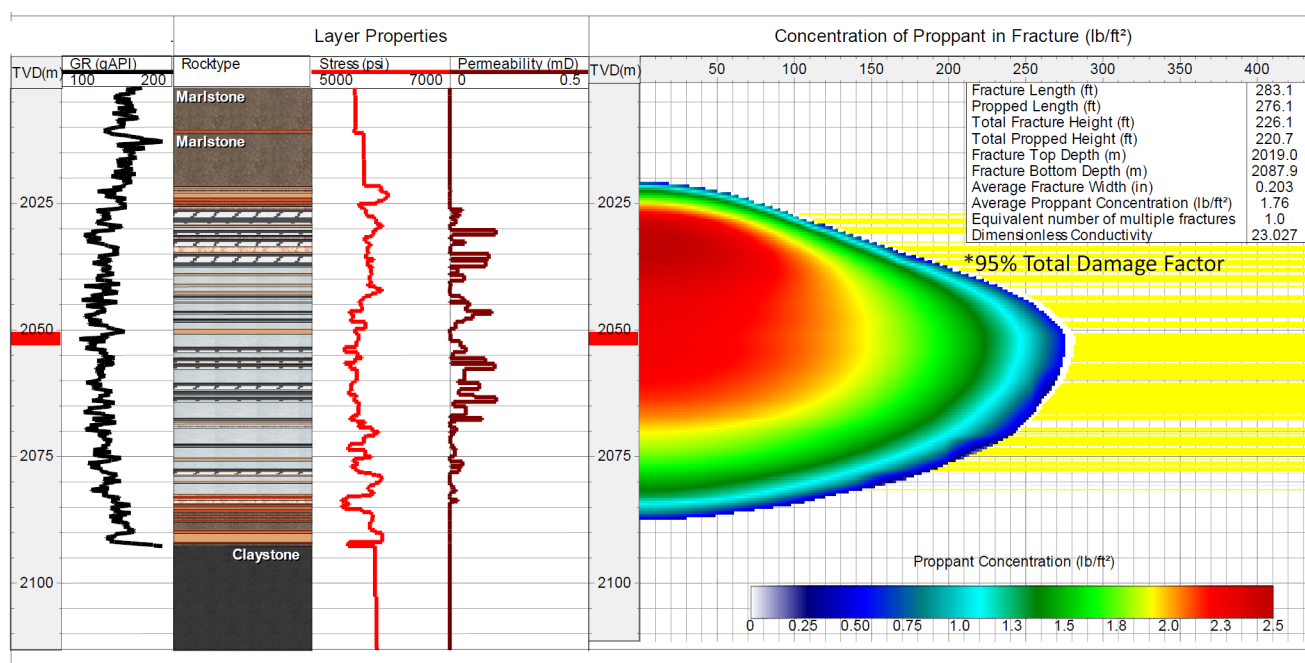


Figure 9—Propped fracture placement results for lower interval high tectonics case.

Table 4—Propped fracturing placement results summary.

Scenario	Zone	Propped Xf	Propped h	Avg Width	Avg Prop Con	FcD @ 95% Damage
()	()	(ft)	(ft)	(in)	(lbs/ft ²)	()
Low Tectonics	Upper	315	222	0.190	1.66	6.2
High Tectonics	Upper	271	255	0.198	1.73	7.1
Low Tectonics	Lower	269	211	0.197	1.71	24.4
High Tectonics	Lower	276	220	0.203	1.76	23.0

Acid Fracturing

The success of acid fracturing will depend heavily on the actual carbonate rock properties. Thorough lab testing with actual core material is recommended to define rock and acid reaction properties and reduce risk. For this initial scoping and design exercise, all available information about the carbonate rock properties from the target formation is considered. Assumptions for rock and acid reaction properties are made based on regional experience in analogue fields and fracture simulator default values, where necessary.

Understanding the rock carbonate and clay content is important for acid fracture modelling. These will affect the acid reaction rate, the amount of material available to react, and the etched fracture conductivity created during pumping and maintained during production. For this exercise, the limestone content of each layer was taken from core X-ray diffraction (XRD) results. For the target intervals consisting of argillaceous limestone, dirty limestone and clean limestone, this data showed a carbonate content ranging from 50 – 85%, a quartz content from 13 – 30% and a clay content (mixed layer/chlorite) from 2 – 30%. Acid solubility tests at room temperature with 5% HCl showed a solubility of 60 – 80%. This gives an indication that there is likely to be sufficient solubility of greater than 70% carbonate across nearly all primary pay intervals, which is generally considered the minimum needed for acid fractures to have a chance of success. If etched channels can be created, the hard, consolidated and layered nature of the limestone may provide excellent etched fracture conductivity that stays open under production conditions. Layers with high clay content are concerning as the clays may be released by the acid resulting in clogging/damaging of the fracture.

Like for propped fracturing, the low and high tectonic stress scenarios were considered for acid fracturing. The uncertainty of the in-situ stress contrast is expected to pose less of a challenge in acid fracturing as these fractures tend to stay more contained due to lower net pressure and the dissolution action of the acid.

Moderate fracture complexity and extra leak-off due to the potential presence of natural fractures and the effects of acid are also modeled, in agreement with the propped fracturing scenarios. Fracture complexity is less concerning for acid fracturing as the acid will tend to reduce fracture complexity through dissolution of carbonate rock.

No entry friction is modelled, as more perforations can be shot and opened by acid that will also work to reduce tortuosity as it dissolves rock. Acid fracturing (with no proppant) allows for longer perforation intervals to be shot which can improve the chances of vertical height coverage during placement (if done correctly), somewhat mitigate risk of poorly stimulated higher clay-content layers acting as vertical flow barriers, and reduce non-Darcy flow effects during production. For this exercise a 10 meter long perforation interval is modelled towards the middle of the interval targeting hard, clean limestone layers. Interval lengths up to 30 meters can be considered, including the use of multiple clusters/intervals per stage if needed. A final perforation interval recommendation will be on a case-by-case basis for any exploration well.

For the pad fluid system, the same guar-based borate crosslinked fluid used for propped fracturing in the area can be used, keeping operations simple. The recommended acid system for this base case treatment design is HCl 20% plus additives. This is to be determined by lab testing, but for acid fracturing, higher concentrations of acid with more dissolving capacity are often desirable to improve the ability to get active acid out to the tip of the fracture. There is also a practical upper limit of 20% HCl for operational reasons in the area. A retarded acid system can be considered but is not the initial recommendation as the BHST is less than 200°F (180-190°F) and due to the presence of hard, tight, low porosity carbonate rock that is expected to have a reaction rate constant on the lower end of the range for carbonates. Straight HCl is also operationally easier to handle/pump/mix and is considerably cheaper than any retarded acid system. Another option to consider is to include extra HCl acid additives to create a mud-and-silt removing acid system for improved clean-up of any dissolved pathways in this type of 'dirty' limestone formation. This type of upgrade to any HCl 20% acid system could be investigated based on lab testing of actual cores/cuttings. Finally, no diverting or leak-off control acid is recommended initially as they may not provide measurable benefit and they will result in higher costs and more complex operations. If significant natural

fractures are present, some leak-off control agent such as degradable fibers or particulate can be included in parts of the pad stages for enhanced leak-off control. Simplicity and cost control are clear drivers.

Treatment design considerations include an acid volume of 350 to 500 gal/ft of treatment height and a pad fluid volume of 200 to 300 gal/ft of treatment height. The fracture height containment, or modelled fracture height, and mineralogy of contacted layers drove the treatment volume ranges.

For the upper interval, the high tectonics case treatment schedule can be seen in Table 5, a treating plot in Figure 10, and acid fracture placement results in Figure 11. The pump rate is limited to 16 bpm and the stage pad volume to 12,500 gal for this case. Smaller volumes and rates are used to target the thinner upper interval when tectonics are high (and stress contrast is low or non-existent) to avoid breaking out of zone. This results in a slightly shorter, lower conductivity fracture. The flush fluid can be 3% KCl with friction reducer. The job size is based on the best technical design to achieve fracture geometry and conductivity goals. There is limited benefit in increasing the job size, however smaller job sizes can be considered based on operational and/or cost considerations, lab testing results and updated reservoir information.

Table 5—Acid fracturing treatment schedule and material totals for upper interval high tectonics case.

Stage #	Stage Type	Elapsed Time min:sec	Fluid Type	Clean Volume (gal)	Slurry Rate (bpm)
1	Main frac pad	18:36	Crosslinked Gel	12,500	16.00
2	Main frac acid	55:48	HCl 20%	25,000	16.00
3	Main frac pad	74:24	Crosslinked Gel	12,500	16.00
4	Main frac acid	111:36	HCl 20%	25,000	16.00
5	Main frac pad	130:12	Crosslinked Gel	12,500	16.00
6	Main frac acid	167:24	HCl 20%	25,000	16.00
7	Main frac overflush	182:17	Linear Gel	10,000	16.00
8	Main frac flush	186:03	KCl 3%	2,535	16.00

Material	Quantity	Units
HCl 20%	1,786	bbbls
Crosslinked Gel	893	bbbls
Linear Gel	238	bbbls
KCl 3%	60	kllbs

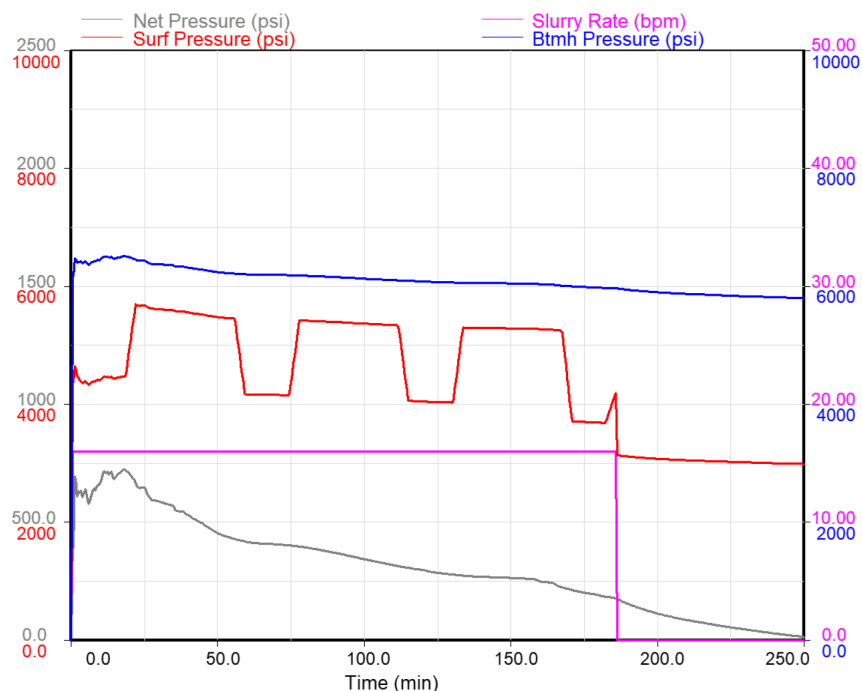


Figure 10—Acid fracture treating plot for upper interval high tectonics case.

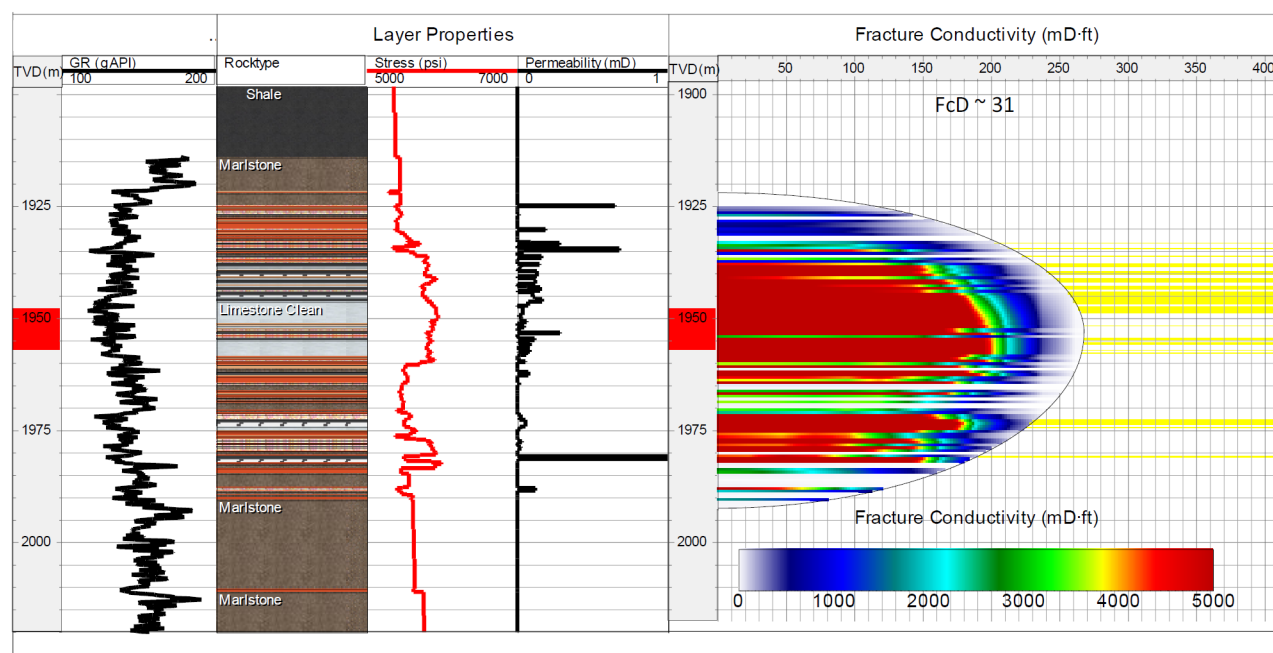


Figure 11—Acid fracture placement results for upper interval high tectonics case.

For the lower interval, the low tectonics case treatment schedule can be seen in Table 6, a treating plot in Figure 12, and acid fracture placement results in Figure 13. The pump rate is up to 24 bpm for this case resulting in a long effective half-length while the fracture still stays well contained. The high expected friction pressure of the acid limited this modelling exercise to 24 bpm. The flush fluid can be 3% KCl with friction reducer. The job size is based on the best technical design to achieve fracture geometry and conductivity goals. There is limited benefit in increasing the job size, however smaller job sizes can be considered based on operational and/or cost considerations, lab testing results and updated reservoir information.

Table 6—Acid fracturing treatment schedule and material totals for lower interval low tectonics case.

Stage #	Stage Type	Elapsed Time min:sec	Fluid Type	Clean Volume (gal)	Slurry Rate (bpm)
1	Main frac pad	19:50	Crosslinked Gel	20,000	24.00
2	Main frac acid	49:36	HCl 20%	30,000	24.00
3	Main frac pad	69:26	Crosslinked Gel	20,000	24.00
4	Main frac acid	99:12	HCl 20%	30,000	24.00
5	Main frac pad	119:02	Crosslinked Gel	20,000	24.00
6	Main frac acid	148:48	HCl 20%	30,000	24.00
7	Main frac overflush	158:43	Linear Gel	10,000	24.00
8	Main frac flush	161:45	KCl 3%	3,047	24.00

Material	Quantity	Units
HCl 20%	2,143	bbls
Crosslinked Gel	1,429	bbls
Linear Gel	238	bbls
KCl 3%	73	klbs

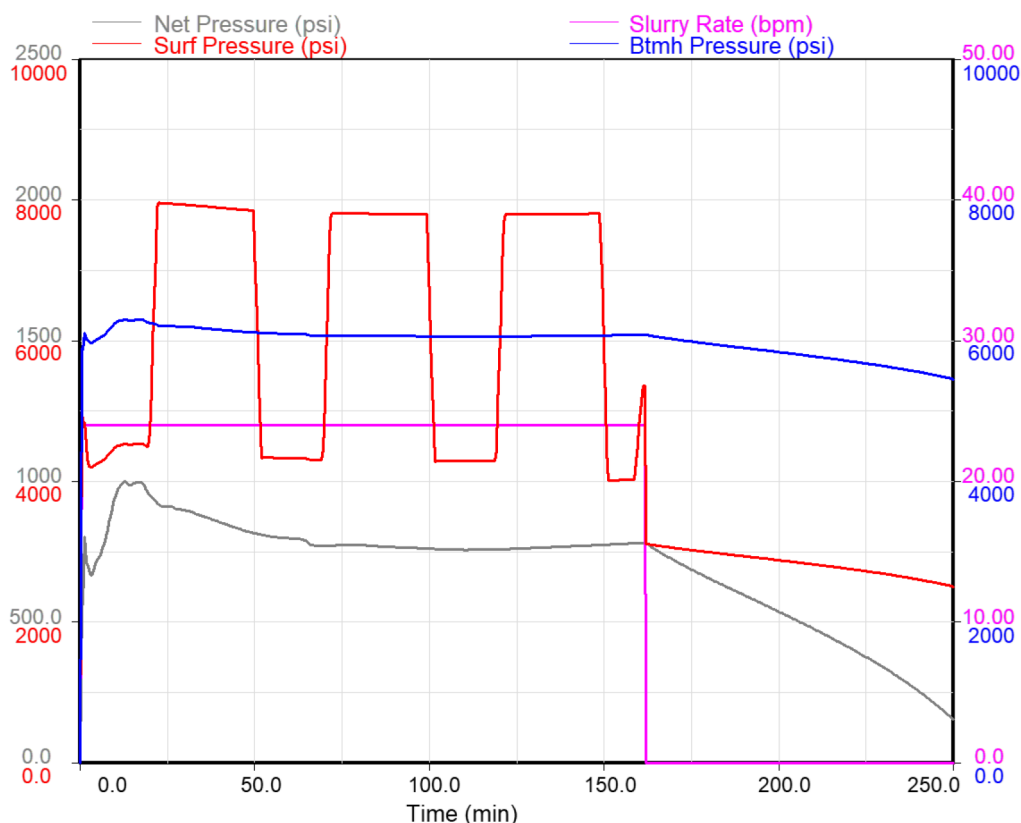


Figure 12—Acid fracture treating plot for lower interval low tectonics case.

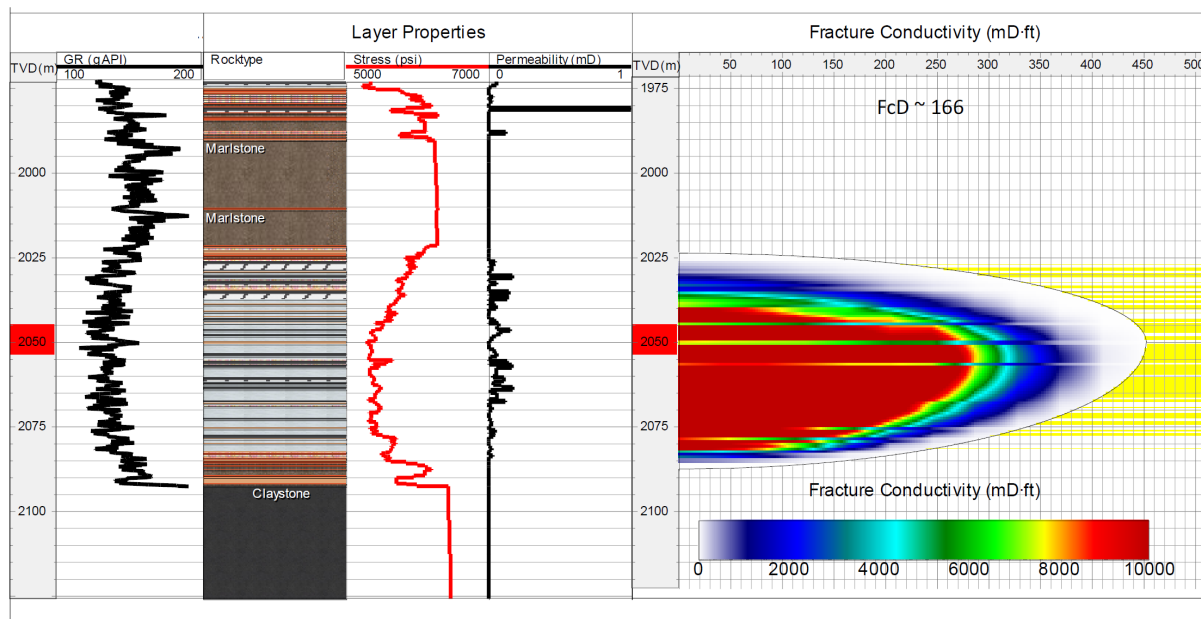


Figure 13—Acid fracture placement results for lower interval low tectonics case.

A summary of the acid fracturing placement results can be seen in Table 7. Based on the acid reaction and lithology inputs in the simulator, there is likely to be sufficient etched fracture conductivity created across all layers for vertical connectivity across entire simulated height. All treatments effectively cover the entire target interval and achieve sufficient fracture conductivity. Sufficient fracture height containment is the bigger concern over zonal height coverage, especially for the high tectonics cases where there is limited

to no stress contrast. Better fracture height containment will result in longer effective etched half-lengths. Design volumes and fluids can be optimized based on lab testing results and actual drilling results of well.

There exists a small risk that some non-reactive (or insufficiently reactive) layers may exist that will not be effectively etched resulting in poor vertical stimulation and flow barriers to vertical flow during production. Lab testing and improved understanding of lithology and acid solubility will be needed to quantify this risk. A longer perf interval may be able to reduce this risk by improving the connection of all layers horizontally back to the wellbore, but needs to be evaluated on a case-by-case basis.

Table 7—Acid fracturing placement results summary.

Scenario	Zone	Etched Xf	Avg Frac Cond	FcD
()	()	(ft)	(mD-ft)	()
Low Tectonics	Upper	323	1,215	24
High Tectonics	Upper	244	1,190	31
Low Tectonics	Lower	419	3,681	166
High Tectonics	Lower	317	3,042	182

In summary, acid fracturing appears to be a good option for stimulation of this tight limestone formation. Good vertical height coverage is achieved in all scenarios, but this is not the key challenge in this reservoir, especially since the plan is to treat it in two stages. Adequate to good etched fracture length is modelled, but this will depend on the containment of the acid fracture treatment. Strong containment will lead to good etched fracture length while poor containment will lead to poor to adequate fracture length. Fracture containment depends on the formation (stress contrast and rock fabric), but treatment design and perforation strategy can play a role. Lower rate and lower pad volume will lead to better fracture containment.

The key risk for acid fracturing is in creating etched fracture conductivity that remains open under production (increased effective stress). Both the ability to create etched fracture conductivity (acid solubility) and have it remain open (clays/impurities/fines) are important. Lab testing will be important to improving this understanding. Local experience with acid washes on the target formation indicate a positive formation response can be achieved.

Matrix Acidizing

As the low range of permeability, and even permeability-height product (kh), of the target formation became evident, the practicality of performing a true matrix acidizing treatment on this target formation was already uncertain. In reality, the success of anything approaching a matrix treatment would probably rely on the ability to connect to a significant open natural fracture network by either cleaning up drilling damage and/or breaking down some perforations. Regardless, the exercise of modelling a matrix acidizing treatment was performed to come up with some optimistic estimates of post-stimulation skin to include in the production forecasting.

Due to the low permeability of the target formation, acid stimulation at true matrix injection rates is expected to be challenging due the low injection rates needed to stay under the minimum in-situ stress in a formation as tight as this target formation. For this exercise, a minimum in-situ stress of 0.80 psi/ft was used as a reference for fracturing pressure, which is slightly below the 0.82 psi/ft range of the lowest stressed layers from the model. The same 0.58 psi/ft pore pressure gradient was used in the software meaning there is only a limited treatment pressure window for acid placement.

The modelling using industry-recognized matrix acidizing design software predicts a treatment injection rate of 0.5 bpm for each of the upper and lower intervals. An injection rate of 1.0 bpm can be assumed if treating both the upper and lower intervals together. This assumes that acid is circulated into position across

the perforations using coiled tubing (recommended) or jointed pipe to avoid any gradual pressure increase associated with significant brine injection prior to reactive acid entering the formation. This also assumes that perforations are open and clean, or an open-hole completion is used. The injection rate may need to start at a lower rate and increase as pressure allows once some acid has entered the formation. The rate should be increased as injection pressure drops throughout the treatment, if possible, for improved diversion effect.

For fluid selection, 15% HCl acid should provide a reasonable balance between reaction rate, dissolution capacity, compatibility and corrosion control under these conditions, but lab testing is recommended to confirm this choice. Organic acids can be investigated for reduced reaction rate/improved worm-holing in this low permeability formations and for improved compatibility with high chlorite clay layers, but this would increase costs and may not provide improved performance. A 3% KCl brine can be used for all injection testing, spacers, over-flushes and other stages requiring a non-reactive fluid. KCl concentration may need to be adjusted to provide stability with actual clays in target intervals. Consider over-flushing the final acid stage, if possible, at least three feet radially away from the wellbore to limit negative effects of any precipitates or solids associated with extended exposure of spent acid on formation. Finally, consider adding mutual solvent or alcohol to improve fluid penetration and clean-up.

Acid diversion and zonal coverage may be challenging as both the upper and lower intervals are expected to have around 40 m of net pay height to target for acidizing. It is recommended to treat both intervals separately to improve zonal coverage of all pay intervals. There is no-to-limited benefit to including diverter stages in the treatment if the pump rate is limited to around 0.5 bpm per interval due to injection pressure throughout the treatment, as the pump time will remain long (around 2 hours or longer if forced to pump slower) while exposing all perforations to acid at high bottomhole pressure close to the closure gradient throughout this long pumping time. In fact, including diverter may cause the treatment to pressure out and/or proceed at an unsustainably low rate due to temporarily sealing off the only better-quality layers that were initially taking fluid. Regardless, for the first exploration well it is worth considering to have a diverter option ready on location in case it is needed if a thief zone develops due to over-stimulation and/or if injectivity is better than expected (high perm layers, open natural fissures). These diverter options would need to be evaluated based operational/technical performance, availability and economics. They are likely to include nitrogen foam (linear gel plus 70% N₂), degradable, multi-modal diverters pills that seal inside perforation tunnels at the formation face (and across small fissures), and/or viscoelastic or self-diverting acid.

An initial treatment schedule is shown in [Table 8](#). This schedule is based on treating around 40 meters (131 feet) of net height per stage using 2,500 gal 15% HCl. This results in around 20 gal/ft of pay or 3.5 feet of radial penetration of acid based on 6% porosity. A treatment volume of 50 gal/ft is the most common starting point for matrix acidizing design, but that is based on a much higher porosity typical in most common matrix acidizing candidates. An over-displacement of 1,500 gal of KCl brine at 3% or higher is recommended following the acid. This equates to 15 gal/ft or almost 3.0 feet of radial penetration. The pumping time at 0.5 bpm is around 3.5 hours, which is reasonable for matrix acidizing in very tight carbonates. Corrosion inhibition of HCl at this BHST below 200°F is not a concern.

Table 8—Matrix acidizing treatment schedule for the upper or lower interval.

Stage #	Stage Type	Fluid Type	Clean Volume (gal)	Cumulative Volume (gal)	Injection Rate (bpm)	Stage Time (min)	Stage Time
1	Main acid	15% HCl	2,500	2,500	00.50	103.5	103.30
2	Overflush	4% KCl	2,000	4,500	00.50	110.8	214:18

A treating plot for this treatment pumped on the lower interval can be seen in [Figure 14](#). This plot shows a maximum surface treating pressure of only 3,000 psi. Note that even injecting at 0.5 bpm, the treatment

nearly pressures out at the end during the brine over-flush. The skin evolution shows an end-of-job total skin of -3.2 (negative), reduced from an assumed initial skin of 1.0 . This can be considered an optimistic or ideal result that assumes efficient wormhole generation even at this low injection rate with HCl and very good/consistent zonal coverage. The upper interval case results in a -3.6 (negative) skin. To be more in line with real field results, treatment results used for production forecasting assume an ideal case of -3 (negative) skin and a more realistic case of -1 (negative) skin. Acid placement and zonal coverage results for the upper and lower treatments can be seen in Figure 11 and Figure 12, respectively. These tend to agree with the previous observation that reasonable diversion is likely to be achieved in this case by holding a high, constant bottomhole injection pressure with straight HCl acid for multiple hours.

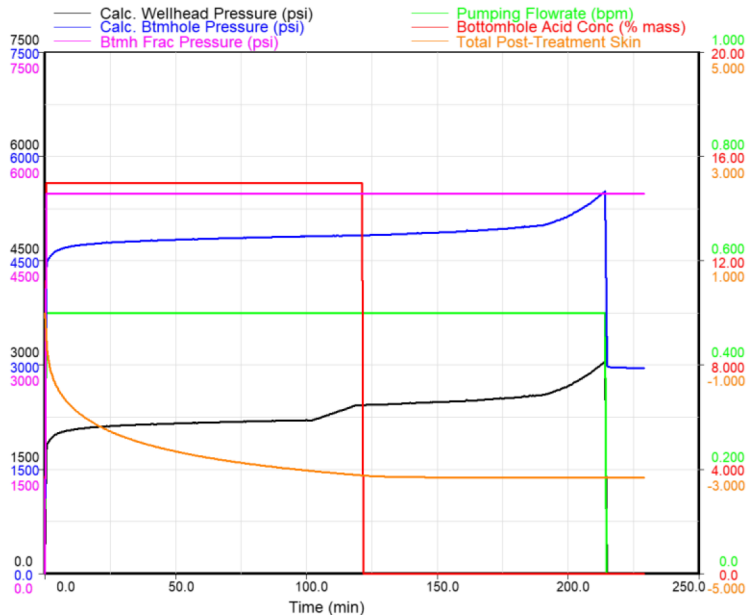


Figure 14—Matrix acidizing treating plot for lower interval.



Figure 15—Matrix acidizing placement results for upper interval.

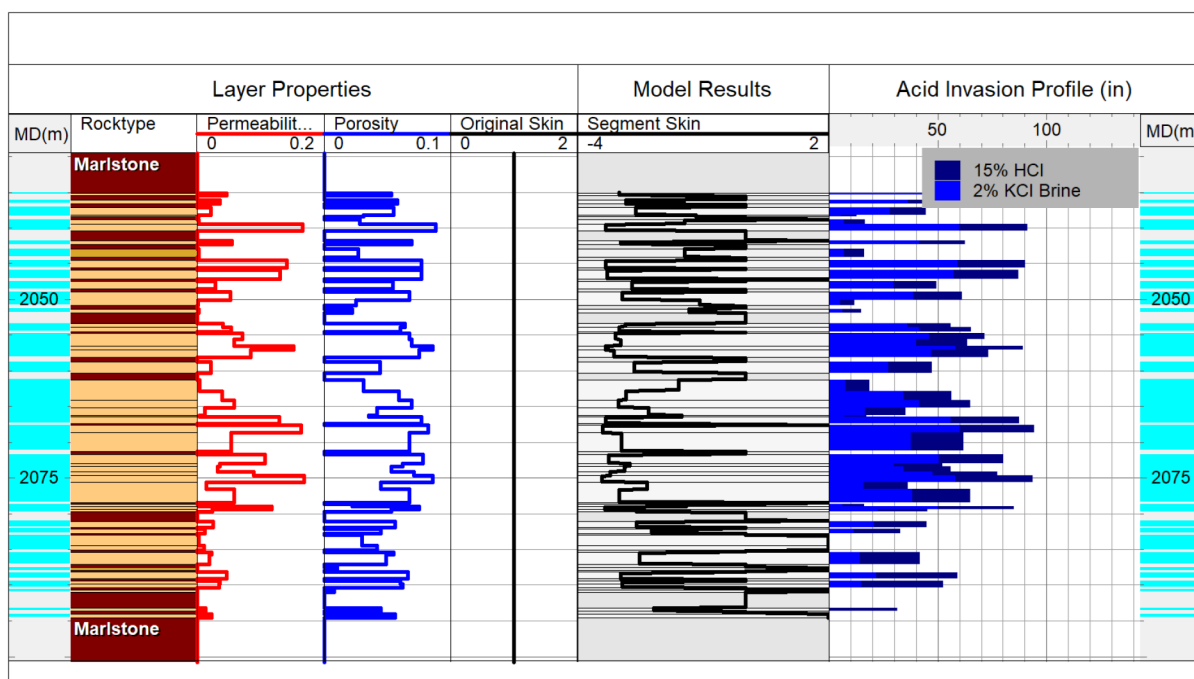


Figure 16—Matrix acidizing results for lower interval.

Based on these modelling results and an understanding of real-world carbonate matrix acidizing results, a post-job skin of -3.0 (negative) is considered the best case, or ideal, outcome. This would need to assume efficient wormholing, even at these limited injection rates with HCl. The wormholes would need to be very stable (perhaps not practical if clay content is high) and extend more than 3 feet out into the formation, beyond the critical matrix. And finally, it assumes excellent diversion and zonal coverage. For production forecasting, the -3.0 skin result is the ideal case.

A more realistic result is to expect a post-treatment skin around -1.0 (negative). Slightly negative skins are common in carbonate acidizing. This result can still exist even with some challenges in wormholing, including inefficient stimulation such as some face dissolution or creation of shorter, conical wormholes due to the low injection rate. Some damage to the wormholes and near-wellbore area may also exist in some layers, especially where higher clay content exists. It also accounts for the reality that not all pay intervals are well stimulated due to challenges with diversion in such long intervals.

There exist some other challenges and considerations in case the matrix acidizing approach is chosen. First, if an open-hole completion is run or the entire interval is perforated, there may not be a contingency fracture treatment option if the matrix approach fails as it may not be practical to even attempt a fracture treatment in such a completion. Second, it is unlikely that matrix acidizing reaches the negative skin necessary to make an economical well in such a tight formation. A well-executed fracture treatment can result in -6 (negative) or lower skin, but -3 (negative) skin is the lowest possible expectation for a matrix treatment. Third, there exists a real possibility that the injection rate achieved is much less than the 0.5 bpm modelled here. This would mean that a more realistic approach is that perforations will need to be broken down and micro-acid fractures created in order to stimulate at any kind of realistic injection rate. This is okay, but will likely lead to poor zonal coverage and a skin closer to the -1.0 value reported above. And finally, if natural fissures are present, an acid treatment targeting the clean-up and connection to the natural fractures may result in similar or slightly better stimulation performance than any matrix acidizing treatment. But even for this case, an acid fracture approach is likely to do this better than a small, low rate acid injection.

Production Forecasting

Two different scenarios of production forecasts were made. One based solely on production from matrix permeability and a second set taking into account potential effects of conductive natural fractures. For the scenarios with only matrix permeability, three cases were run to cover a range of permeability and porosity values (best, base, worst). For the scenarios with natural fracture effects, three cases were run to cover a range of improvement for the effective permeability (compared to the matrix permeability) due to the presence of natural fractures (best, base worst). The effects of natural fractures were modeled using the base case matrix permeability and porosity values. For each case within each scenario, production forecasts were made for all three stimulation types (matrix acidizing, acid fracturing, propped fracturing).

Modeling Effects of Natural Fractures on Effective Permeability

Currently available data on the target formation suggests that natural fractures can be present, but no information on fracture permeability is available. A value for fracture porosity has been estimated from the log data from the offset well, and the average value of the fracture porosity is in the range of 0.5 % to 1.0 %. This estimated fracture porosity is much higher than the value estimated from the log data of the other well studied, which was 0.1%.

Due to the lack of data and the large uncertainties involved, the production forecasting for the target formation for the stimulation study has been performed with the simple single-layer, single-porosity reservoir simulation model in a commercial simulation model (Crocket, 1986). This type of simple modeling is most applicable for this type of prospect evaluation. The effect of natural fracture permeability on effective (upscaled) reservoir permeability is a very complicated issue, dependent on many different factors, such as fracture permeability, fracture orientation, fracture spacing, and fracture extent, among others. Since none of this data is available for the target formation, we have decided to use a simple approach, based on data from other limestone formations with natural fractures and comparable porosities. One such formation is the Ellenberger formation in the USA, where useful data is available from an old publication (Kelton, 1950) shown in Table 9. The ratio of the last two numbers in the table gives an estimate of the potential uplift in the effective permeability due to natural fractures. In this set of four cores, this "uplift" factor is 2, 6, 33 and 10. We have decided to ignore the extreme value of 33 as an outlier and take a range of 2 – 10 for increase in effective permeability for this type of low permeability limestone. We therefore have 3 scenarios for permeability enhancement due to natural fractures which are shown in Table 10.

Table 9—Effective vs. matrix permeability of Ellenberger core (Kelton, 1950).

Matrix vs. Whole Core Data, Ellenburger, Fullerton Field					
Group		1	2	3	4
Gas,	% bulk	0.08	0.91	0.63	3.72
Oil,	% bulk	0.06	0.06	0.88	1.84
Water,	% bulk	2.07	1.65	1.66	3.27
	% pore	94	63	52	39
Matrix porosity,	% bulk	1.98	1.58	2.56	7.92
	% total pore	90	60	81	94
Total porosity,	% bulk	2.21	2.62	3.17	8.40
K_{max} ,	md	10	409	23	94
K_{90° ,	md	0.6	1.2	10	38
Matrix K ,	md	0.3	0.2	0.3	3.7

Table 10—Scenarios for effect of natural fractures on matrix permeability.

Natural Fracture Effect Scenario	Multiplier on Matrix Permeability
Low	2
Mid	5
High	10

Reservoir Parameters for Single Layer Models

As described in the previous section, three different scenarios have been defined for the average porosity and permeability for the single layer reservoir simulation model. These are shown in Table 11. The base and worst cases were created by reducing the porosity from the best case by 1/3 and 2/3 and calculating the permeability for these porosity values using the poro-perm relationship from the synthetic log. As expected for this kind of tight reservoir, the permeability is drastically reduced for slightly lower porosity. For the worst case scenario, the permeability is in the range of an "unconventional" reservoir. Even the base case scenario is a "Tight Gas" reservoir. Based on this range of matrix permeability, it is clear that hydraulic fracturing will be necessary to make an economic well. Only in the case of an enhanced permeability due to natural fractures could matrix acidizing give a significant post-stimulation gas rate.

In addition to the permeability and porosity, the other parameters needed for the single well model used for production forecasting are the net thickness, water saturation and drainage area. The drainage area has been assumed to be the same for all cases at this stage. A rectangular drainage area of 5000×5000 ft has been used (574 acres). This may be optimistic for the worst case scenario, but due to the low permeability, the drainage area is not really limiting the production for the 5 year forecasts that were made. The additional reservoir parameters are shown in Table 12. The total GIIP for both zones range from about 15 BCF for the worst case to 43 BCF for the best case. These GIIP numbers are relatively large, due to the assumed well spacing, but the recovery factors based on this drainage area will be low, due to the low permeability. The more important parameter is the 5 year recovery, rather than the GIIP. For a real field development of such a low permeability reservoir, the well spacing would most likely be much smaller, improving the recovery factor but requiring more wells. In a study of the long-term production from a well-known Tight Gas well that was part of the Gas Research Institute research program (SFE #3), with permeability in the range of 0.005 to 0.01 mD, the initial estimate of the drainage area was 160 acres (Shaoul, 2016). Based on long-term production of 20 years, the actual drainage area was estimated to be closer to 80 acres, most likely because of subsequent infill drilling on a 60 acre spacing.

The production forecasts are performed using a commercial single-layer, single-phase 2D reservoir simulator that explicitly models the propped or acid fracture, including the effects of effective stress over time for both propped and acid fractures. This effect is important for acid fractures, which will lose fracture conductivity as the effective stress increases due to depletion over time.

Table 11—Scenarios for reservoir parameters for production forecasting.

	Best Case	Base Case	Worst Case
Upper Zone - Avg Porosity (%)	7.2	4.8	2.4
Upper Zone - Avg Perm (mD)	0.16	0.029	0.0016
Lower Zone - Avg Porosity (%)	5.7	3.8	1.9
Lower Zone - Avg Perm (mD)	0.053	0.0097	0.0005

Table 12—Net thickness, Sw and GIIP for production forecasting.

	Best Case	Base Case	Worst Case
Upper Zone			
Porosity (%)	7.2	4.8	2.4
Sw (%)	40	40	40
Net Pay Thickness (m)	17.6	17.6	17.6
GIIP (BCF)	13.5	9.2	4.6
Lower Zone			
Porosity (%)	5.7	3.8	1.9
Sw (%)	33	33	33
Net Pay Thickness (m)	41.3	41.3	41.3
GIIP (BCF)	29.3	19.4	9.7

The properties of the propped and acid fracture are imported automatically into the reservoir simulator and averaged over the pay zone height along the fracture length, which is divided into 10 segments to model the decreasing fracture conductivity moving away from the wellbore. The model can also handle non-Darcy flow in the fracture, but that is not an issue with the low gas rates in this kind of reservoir. For the matrix acidizing cases, the negative skin is translated into an effective wellbore radius in the reservoir simulator.

Production Forecasting Constraints

To perform the production forecasts for the initial 5 years of production, a maximum gas rate for the well of 15 MMscf/day was assumed, based on the expected production tubing size that would be used. Since upper and lower zones are simulated separately, this has been split evenly between the two zones, with each zone given a maximum rate of 7.5 MMscf/day. The pressure constraint is a minimum bottomhole flowing pressure of 1000 psi, based on a realistic surface pressure for the likely gas gathering system. Since a single layer model is used, each zone is treated as a separate well but by using a bottomhole pressure constraint, a realistic commingled production forecast can be made by adding the individual gas rates together. A time horizon of 5 years was chosen to evaluate the project economics, but in such tight gas reservoirs, a long tail of low rate production can continue for 10-20 years, if the production tubing is small enough to avoid liquid loading.

Production Forecasting Results

Production forecasts were run for the three different reservoir quality scenarios for both acid and proppant fracturing. For the base case reservoir quality, an additional sensitivity was run for the three different natural fracture effects for both acid and proppant fracturing. For matrix acidizing, only the base case reservoir quality was used, with two different cases for negative skin (-1 and -3) as well as an upside case with a skin of -3 and the benefit of natural fractures for the base case reservoir quality. All the results are summarized in Table 13. For the worst case reservoir quality, no stimulation is able to deliver a reasonable gas rate and 5 year recovery. For the base case reservoir quality, we can see that the 5 year recovery varies from 0.7 to 1.0 BCF for matrix acidizing and 1.6 to 1.9 BCF for propped and acid fracturing, respectively. These are also very marginal sub-economic numbers for a well in this environment. It is only for the best-case reservoir properties, or the cases with natural fracture enhanced permeability that some kind of economic recovery could be achieved, on the order of 5 BCF or more.

Table 13—Results of production forecasting.

Stimulation Type		RQ Case	NF Case	Gas Rate @ 5-Days (MMscf/d)	Gas Rate @ 6-Months (MMscf/d)	Cum Gas @ 5-Years (BCF)
Propped Fracturing		Best	None	9.0	4.5	6.9
		Base	None	2.5	1.2	1.6
		Worst	None	0.2	0.07	0.1
		Base	Best	10	6.0	7.5
		Base	Base	7.0	3.5	4.9
		Base	Worst	4.0	1.6	2.4
Acid Fracturing		Best	None	14.5	6.0	8.1
		Base	None	4.0	1.3	1.9
		Worst	None	0.6	0.2	0.2
		Base	Best	15	9.0	10.5
		Base	Base	15	5.0	6.8
		Base	Worst	7.0	2.3	3.3
Matrix Acidizing	-3 Skin	Base	None	1.0	0.6	1.0
	-1 Skin	Base	None	0.6	0.4	0.7
	-3 Skin	Base	Best	7.0	4.7	6.6
	-3 Skin	Base	Base	4.4	2.9	4.5

The production forecasts all show the typical rapid decline of a tight-gas well. The initial production from the acid fracture cases is higher, due to the higher fracture conductivity. But this leads to a faster decrease in gas rate over time, so this is purely an acceleration effect. Even though the acid fracture conductivity is modeled as dropping over time by a factor of 8, the fracture conductivity is still effectively infinite, due to the low reservoir permeability. For the propped fractures, the fracture conductivity is not infinite, leading to some pressure drop in the fracture at early time. A comparison of all three stimulation methods for the base case reservoir permeability is shown in Figure 17. Fracture stimulation clearly outperforms even the most optimistic matrix stimulation in this situation.

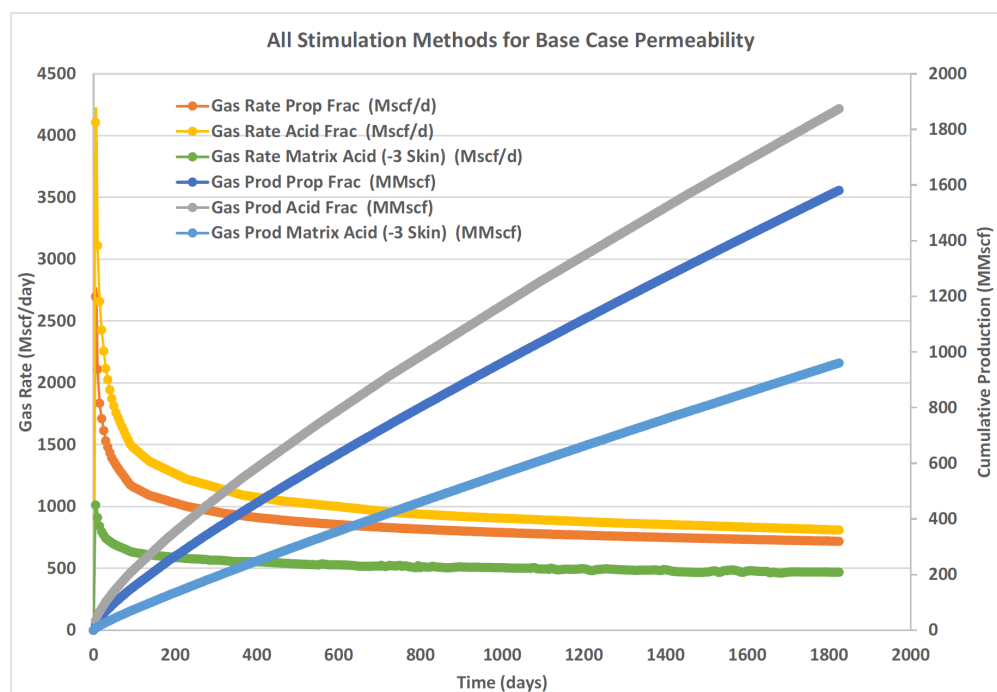


Figure 17—Comparison of all Stimulation Methods for base case reservoir quality.

The effect of reservoir quality for the acid fracture production forecasts is shown in Figure 18. The worst-case scenario production is hardly visible on the plot. The difference between the base case and the best

case is also quite significant, a factor 4 larger recovery. The sensitivity on potential upside from natural fractures in combination with the base case reservoir quality is shown in Figure 19. The benefit of natural fractures is very significant in such a tight reservoir. The presence of widely distributed natural fractures that translate to a higher effective reservoir permeability would have a huge economic benefit in such a tight reservoir, potentially increasing the 5 year recovery from a marginal 2 BCF to something in the range of 4-10 BCF, resulting in an economic well.

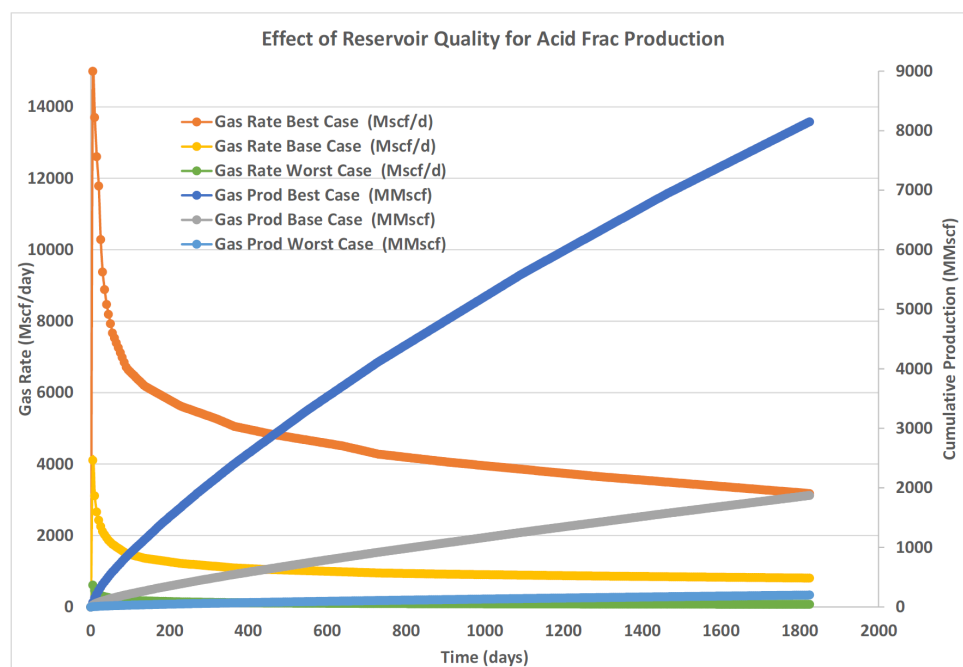


Figure 18—Comparison of acid frac production for different reservoir quality scenarios.

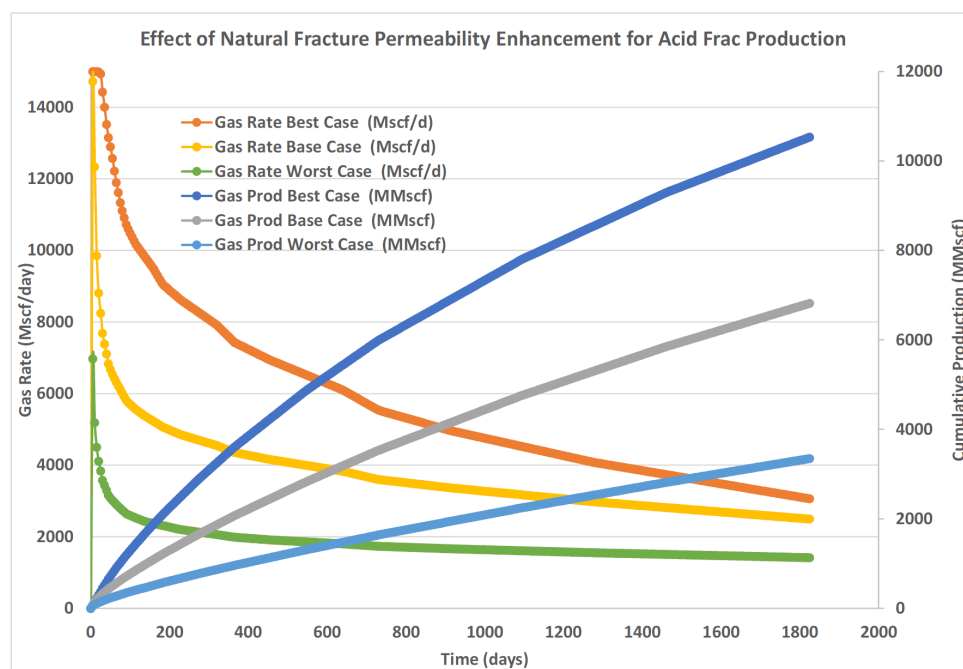


Figure 19—Sensitivity on natural fracture enhancement of matrix permeability for acid fracture.

Completion Options

The decision on the well completion will be important for the stimulation, production and economic success of an exploration well in the target formation. No final completion selection will take place until an exploration well is planned. However, it is important to understand what will be needed from a completion for an exploration well and to discuss benefits and drawbacks of various options ahead of time.

The lowest cost and simplest option is an open-hole completion with a slotted liner run across the reservoir section. The entire open-hole reservoir section could then be stimulated via matrix acidizing. If matrix acidizing is the stimulation method of choice, this is an adequate completion choice as it is relatively cost effective and low risk, while still allowing an effective matrix stimulation treatment to be pumped. However, production forecasting indicates that the most likely stimulation method is either acid fracturing or proppant fracturing owing to the expected low permeability of the reservoir target. The downside of this open-hole completion option with a slotted liner is that it is not suitable for acid or proppant fracturing.

Any suitable completion option for the first exploration well in this target formation should allow for proppant and acid fracturing. The decision on the final stimulation method (acid or proppant) may not be made until the well is drilled, cored and logged. Lessons learned from the first well can then be applied to future well completions, especially once the stimulation method is finalized moving forwards.

The other criteria for any completion option is that multi-stage fracturing (MSF) needs to be possible in the completion, as two fracture stages are expected to be needed to treat both the upper and lower targets separately since the gross height of the entire interval can exceed 200 m and a 30 to 50 meter marlstone/claystone/shale section exists in between upper and lower pay intervals.

As a starting point, the well and tubular design can be based on the wells currently being proppant fractured in the area in an under-lying reservoir that is also tectonically active. These wells are completed with a 7-inch cemented casing or liner with short perforated intervals of around 3 meters. If acid fracturing is chosen, there is more flexibility for the length, location and number of perforated intervals. The cemented casing/liner provides the best and most certain zonal isolation for any fracture stage.

Treatments are pumped down either a 4.5 or 3.5-inch tubing/DST string. Since this project target formation is shallower than the formations currently being fractured in the area and treatment designs are for comparable pump rates, from 16 – 24 bpm, the tubular friction and closure stress should be lower resulting in even more manageable surface treating pressure. A smaller 3.5-inch tubing may be a better option as it may make post-stimulation clean-up and DST/well testing operations more practical for what are expected to be relatively low gas rates.

Proppant plugs are a practical and cost-effective option for zonal isolation between frac stages. For propped fracturing, the option is straight forward as there is already proppant and proppant handling equipment on location. For acid fracturing, a proppant plug can be placed with coiled tubing across the perforations of the lower interval after it has been clean-up and tested. The proppant plug can then be cleaned out with coiled tubing after the upper interval has been stimulated, cleaned up and tested. Proppant plugs are a proven option worldwide for zonal isolation and are relatively low risk.

The other option to consider is to change to a monobore completion (4.5" casing to surface, or 4.5" liner tied back to surface with 4.5" tubing) as it is a very good option for MSF completions. While MSF completions have many potential benefits, before one is chosen a review will have to be performed to ensure the option is operationally, technically and economically feasible for this application. Two key advantages exist for this option. First, it allows for larger casing perforation guns to be used for perforating all intervals leading to improved connection between wellbore and formation for all stages. This may help mitigate issues with multiple fractures and fracture complexity. Finally, it can allow for easier/faster/cheaper zonal isolation through the running of a frac plug between frac stages for zonal isolation. Plugs can be milled with coiled tubing or jointed pipe post-fracturing rather than cleaning out proppant.

If acid fracturing is the stimulation method of choice, there are some additional considerations/options to be considered. First, an open-hole completion can be considered. Acid will mitigate issues with initiating a fracture without perforations in the open hole and reducing effects of multiple fractures and fracture complexity. And there are no longer any proppant admittance concerns. Next, acid fracturing would safely allow for an MSF lower completion with balls and sleeves, or even coiled tubing activated sleeves, to be considered for faster/simpler operations and reduced costs. This MSF completion could use open-hole packers for zonal isolation outside the liner. Some leakage across open-hole packers when acid fracturing can be expected, but this can be mitigated by using multiple packer elements included above and below target intervals. Degradable (or millable) balls can be dropped from surface to shift open sleeves across frac ports and provide zonal isolation between frac stages inside the liner.

Other well completion and post-fracture procedures need to be considered prior to the first exploration well. In this area, a pre-fracture drill stem test (DST) is typically performed to validate the decision to fracture and for improved treatment evaluation. In this tight target formation, a pre-fracture DST may be challenging or not practically possible. A simple acid wash may help to achieve a successful pre-frac DST, especially if it can help connect to some natural fractures. Also, perforating shorter intervals for fracturing may limit options for a pre-frac DST which typically consists of testing the entire pay interval. Perforating for acid fracturing is more flexible than propped fracturing, allowing longer perforated intervals for any pre-fracture DST. Most parts of the world do not perform pre-stimulation DSTs in comparable 'unconventional' or ultra-tight plays.

Finally, coiled tubing is likely to be necessary to maintain operational flexibility. It can be used for placing, adjusting and cleaning out proppant plugs. It will also facilitate a nitrogen (N₂) kick-off to initiate or improve post-stimulation clean-up and/or for DSTs. Placement of any acid wash or matrix treatment is greatly improved, along with zonal coverage, through the use of coiled tubing. Ultimately, coiled tubing can also be used for the milling of any balls and/or ball seats. The availability of coiled tubing for this type of well operation will provide more flexibility.

Conclusions

- For extremely low permeability carbonate reservoirs, matrix acidizing cannot compete with acid or proppant fracturing, unless there is significant permeability enhancement due to the presence of open natural fractures.
- For deep, high modulus carbonate formations, propped fracturing is very likely to be difficult, with a high chance of problems placing proppant. In such a situation, the most promising stimulation method to try first, in an exploration well, is acid fracturing. The only exception to this recommendation is if there is core data available that shows that the formation cannot be successfully etched with acid due to low reaction rate at reservoir conditions or the presence of non-reactive components, including clays, in the mineralogy.
- If it is possible to create an acid fracture in this reservoir, the dimensionless fracture conductivity will effectively be infinite, due to the very low matrix permeability, even with the most conservative estimate of acid fracture conductivity. With a propped fracture, it is very likely that the fracture will be conductivity limited, due to the difficulty of placing a high proppant concentration and the difficulty of cleaning up the proppant pack in a low permeability reservoir.
- In this exploration prospect, in the worst case reservoir quality scenario, it was not possible to create an economic well with any type of stimulation. For the base-case reservoir quality scenario, the economics are borderline, unless there is some permeability enhancement from natural fracturing. In the best case reservoir quality scenario, an economic well is possible with any of the stimulation methods.

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